

PSO Task: Rescue Drone Charging Station Location

Course: Evolutionary Algorithms

Problem Statement

A city wants to install a **rescue drone charging station**. The station should be placed at a location that is close to important emergency demand points such as hospitals, rescue centers, residential areas, and public locations.

The station location is represented as:

$$X = [x, y]$$

where:

x = horizontal location in km

y = vertical location in km

The search area is:

$$0 \leq x \leq 10$$

$$0 \leq y \leq 10$$

Emergency Demand Points

Point	Description	x_k	y_k	Weight w_k
A	Hospital	2	8	4
B	Rescue center	8	7	3
C	Residential area	3	3	2
D	Main road junction	7	2	2
E	University area	5	5	1

A higher weight means that the charging station should preferably be closer to that point.

Objective Function

The weighted distance cost is:

$$J_1 = \sum_{k=1}^5 w_k [(x - x_k)^2 + (y - y_k)^2]$$

There is also a restricted circular area where the station is not allowed.
The center of the restricted area is:

$$(x_c, y_c) = (5, 5)$$

and the radius is:

$$R = 1$$

The distance of the charging station from the restricted-area center is:

$$d = \sqrt{(x - x_c)^2 + (y - y_c)^2}$$

If the station is inside the restricted area, apply the following penalty:

$$\text{Penalty} = 1000 \times \max(0, R - d)^2$$

The final objective function is:

$$J = J_1 + \text{Penalty}$$

The goal is:

Minimize J

Part A: One PSO Iteration by Hand

Perform **one PSO iteration by hand** using a small swarm of your own choice.

You may choose:

swarm size = 3 or 4

For each particle, choose initial values of:

$$x, \quad y, \quad v_x, \quad v_y$$

within the allowed bounds. You may take the initial velocities as zero for simplicity.
Use the following PSO parameters for the hand calculation:

$$w = 0.7, \quad c_1 = 1.5, \quad c_2 = 2.0$$

For the first iteration, assume:

$$pbest_i = x_i$$

because each particle has not moved yet.

In your hand calculation, show the following steps:

1. Select your initial swarm.
2. Evaluate the objective value J for each particle.
3. Identify the initial global best, $gbest$.
4. Choose your own random values of r_1 and r_2 in the range $[0, 1]$.
5. Update the velocity of each particle using the PSO velocity equation discussed in class.
6. Update the position of each particle.
7. Evaluate the new objective values.
8. Update $pbest$ and $gbest$ if improvement occurs.

Clearly show all random numbers used in your hand calculation.

Part B: Python Implementation

Write a Python program using PSO to solve the same rescue drone charging station location problem.

You may use the **DEAP library**.

You are free to choose:

swarm size

number of iterations

However, your selected values should be reasonable and must be mentioned in your report.

Use the following PSO parameters:

$$w = 0.7, \quad c_1 = 1.5, \quad c_2 = 2.0$$

Use the variable bounds:

$$0 \leq x, y \leq 10$$

You may choose suitable velocity limits, for example:

$$-2 \leq v_x, v_y \leq 2$$

Required Output

Your program should report:

Quantity	Value
Chosen swarm size	
Chosen number of iterations	
Best x location	
Best y location	
Best objective value J	
Penalty value	

Also plot:

1. Best objective value versus iteration
2. Best x and y values versus iteration

Questions to Answer

1. What does each particle represent in this problem?
2. What is the meaning of $pbest$?
3. What is the meaning of $gbest$?
4. Why is a penalty term added to the objective function?
5. What happens if a particle enters the restricted circular area?
6. What is the role of inertia weight w ?
7. What is the difference between the cognitive term and the social term?
8. What swarm size and number of iterations did you choose? Why?
9. Did the final solution lie inside or outside the restricted area?
10. Why may different students obtain different final answers?

Submission Requirements

Submit the following:

1. Hand calculation of one PSO iteration
2. Python code
3. Screenshot or copy of program output
4. Convergence plots
5. Final best solution
6. Answers to the questions